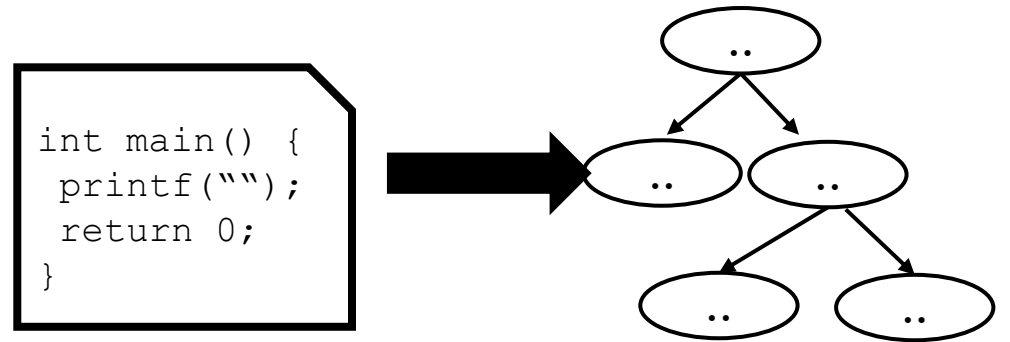


CSE110A: Compilers



Topics:

First & Follow Sets and
Warshall's Transitive Closure

Warshall's Transitive Closure

```
def warshall_full_closure(relation, non_terminals, all_symbols):  
    closure = relation.copy()  
    for k in non_terminals:  
        for i in non_terminals:  
            for j in all_symbols:  
                if closure.loc[i, k] and closure.loc[k, j]:  
                    closure.loc[i, j] = True  
    return closure
```

Understanding Ideas Behind First & Follow Sets Using Transitive Closure

```
grammar = {  
  "Expr" : [  
    ["Unit", "Expr2"]  
  ],  
  "Expr2": [  
    ["Op", "Unit", "Expr2"],  
    []  
  ],  
  "Unit" : [  
    ["(", "Expr", ")"],  
    ["ID"]  
  ],  
  "Op" : [  
    ["+"],  
    ["-"],  
    ["*"],  
    ["/"]  
  ]  
}
```

First Set of Relations : One Production at a time

	Unit	Expr2	Op	Expr	)	-	/	(	*	ID	+
Unit								v		v	
Expr2			v								
Op						v	v		v		v
Expr	v										

First Set : After Warshall's Transitive Closure

	Unit	Expr2	Op	Expr	)	-	/	(	*	ID	+
Unit								v		v	
Expr2			v			v	v		v		v
Op						v	v		v		v
Expr	v							v		v	

Note: FIRST Set calculation, is more complex than Transitive Closure